# Emil Ismayilzada

ismailzadeh.emil@gmail.com | +36 30 156 7713
linkedin.com/in/ismayilzada | github.com/iamemil

## Summary

AI-focused software engineer at BMW Group, building production LLM-powered systems: a LangChain-based agent with tool calling, RAG over production data, and natural-language-to-SQL on top of PostgreSQL and Snowflake. Also builds consumer AI products end-to-end, including **PocketBuddy**, a shipped iOS app with an LLM-based money assistant serving real users. Comfortable across Python, TypeScript/React, and modern data infrastructure.

## Languages & Technologies

| | |
|---|---|
| **LLM & AI** | LangChain, OpenAI APIs, RAG architectures, pgvector, vector embeddings, prompt engineering, tool/function calling, semantic memory, STT/TTS |
| **Languages** | Python, TypeScript, JavaScript, SQL, PL/SQL, C#, Dart |
| **Data & Infra** | PostgreSQL, MongoDB, Snowflake, Apache Kafka, Redis, AWS (QuickSight, S3, Lambda, EC2, ECR, Athena) |
| **Web & Mobile** | React, Three.js, Node.js, ASP.NET, Flutter, iOS |
| **ML & Analytics** | PyTorch, scikit-learn, pandas |
| **Tools** | Git, Docker, Grafana, Microsoft Azure, Microsoft Power Apps |

## Work Experience

**BMW Group — Data Analyst / Software Engineer (AI & Data Platforms)** **Oct 2024 – Present**
*Debrecen, Hungary*

- Built and shipped a **LangChain**-based AI agent with tool-calling that lets operators query production systems in natural language, integrating OpenAI models, real-time token streaming, voice input/output (STT/TTS), and automatic chart generation — adopted as a primary interface for energy module production data.
- Implemented **RAG**-style semantic memory using pgvector and OpenAI embeddings; **40% of queries** now served from cached memory, reducing redundant LLM calls and response latency.
- Designed a **natural-language-to-SQL** pipeline translating user questions into validated PostgreSQL queries against production and Snowflake data, with schema-aware grounding to prevent hallucinated table or column names.
- Built and operate **19 ETL pipelines** ingesting static files, Kafka topics, and Snowflake data into PostgreSQL; reduced dashboard query latency from **30 s to 100 ms** (≈300×) using short-term Redis caching, with Redis-backed uptime monitoring and Grafana alerting.
- Architected the **Digital Twin** for Energy Module Production platform (React, Three.js, Node.js) — interactive 3D production floor, 20+ real-time KPIs, embedded AWS QuickSight dashboards, and multi-language support used by management and shop-floor associates.
- Built an internal admin platform with role-based access control (RBAC), KPI registry, and an **AI memory** management UI with embedding visualization.
- **Mentor** junior engineers and interns on the team's AI/data tech stack through structured onboarding support, custom training materials, pair programming sessions, and task prioritization guidance.
- Volunteered to train management on enterprise AI tools (e.g., Microsoft Copilot), going beyond slide presentations to work individually with each manager until they could use the tools confidently.
- **Drove AI adoption** across the team by tracking Copilot usage progress and coordinating directly with central steering teams to align on rollout strategy and resolve adoption blockers.

**BMW Group — Software Engineer Intern** **Nov 2023 – Jun 2024**
*Debrecen, Hungary*

- Built 2 internal business applications with Microsoft Power Apps, automating daily operational workflows used by **20–40 managers** and reducing manual handoff time; integrated with SharePoint and Excel data sources.

**VisitMe — Software Engineer Intern** **Apr 2021 – Jun 2024**
*Debrecen, Hungary*

- Built and shipped 3 production web applications using React, ASP.NET, and C#, including a guest-list management app that served **600–800 attendees per event** and cut on-site check-in time.

## Projects

**PocketBuddy** — AI & Full-Stack Engineer (Founder) **Live on the App Store**
pocketbuddy.emlsm.tech

- Shipped personal finance app for iOS with an LLM-based money assistant: chat over real transaction data, AI-generated

"Can I Afford This?" verdicts using active budget context, AI daily summary stories, **semantic retrieval**, and a home-screen widget. Subscription-based.

**Rentodo** — Flutter Developer (Founder) github.com/iamemil/rentodo

- Peer-to-peer item rental app built with Dart, Flutter, Firebase Firestore, and Cloud Functions; supports real-time listings, in-app messaging, and push notifications.

**Fisk** — Full-Stack Developer

- Expense tracker (React, ASP.NET, MSSQL) with Receipt capture with QR-codes, auto-categorization, **crowdsourced data**, and merchant data via a public tax service API.

## Education

**University of Debrecen** — Computer Science, MSc **Sep 2022 – Jun 2024**
*Debrecen, Hungary | GPA: 4.25 / 5.0*

**University of Debrecen** — Computer Science, BSc **Sep 2019 – Jun 2022**
*Debrecen, Hungary | GPA: 4.42 / 5.0*